LAKSHYA
JEE 2027

Lecture No.- 27 **MATHEMATICS**

RELATIONS & FUNCTIONS

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Topics *to be covered*

1. Questions on Even & Odd Functions
2. Periodic Functions
3. Question Practice
4.

Question Practice

Question [JEE (Adv.)-2025]

Let $\mathbb{N}$ denote the set of all natural numbers, and $\mathbb{Z}$ denote the set of all integers. Consider the functions $f : \mathrm{N} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathrm{N}$ defined by

$$f(n) = \begin{cases} \frac{(n+1)}{2} & \text{if } n \text{ is odd,} \\ \frac{(4-n)}{2} & \text{if } n \text{ is even,} \end{cases} \quad \text{and} \quad g(n) = \begin{cases} 3+2n & \text{if } n \geq 0, \\ -2n & \text{if } n < 0. \end{cases}$$

Define $(g \circ f)(n) = g(f(n))$ for all $n \in \mathbb{N}$, and $(f \circ g)(n) = f(g(n))$ for all $n \in \mathbb{Z}$. Then which of the following statements is (are) TRUE?

A. $g \circ f$ is NOT one-one and $g \circ f$ is NOT onto

B. $f \circ g$ is NOT one-one but $f \circ g$ is onto

C. g is one-one and g is onto

D. f is NOT one-one but f is onto

Question [IIT-JEE-2005]

If X and Y are two non-empty sets where $f: X \rightarrow Y$, is function is defined such that $f(C) = \{f(x): x \in C\}$ for $C \subseteq X$ and $f^{-1}(D) = \{x: f(x) \in D\}$ for $D \subseteq Y$, for any $A \subseteq Y$ and $B \subseteq Y$, then

HW

(A) $f^{-1}\{f(A)\} = A$

(B) $f^{-1}\{f(A)\} = A$, only if $f(X) = Y$

(C) $f\{f^{-1}(B)\} = B$, only if $B \subseteq f(x)$

(D) $f\{f^{-1}(B)\} = B$

LAST CLASS REVISION

Answer the following Questions:

1. Any straight line with slope (–1) is inverse of itself. → T

2. If we interchange x & y, and functional expression remains same then function is self inverse (if exist) and domain & range are equal sets. → T

3. $f:[-2,4] \to R$ & $f(x) = \frac{3x^4+2x^2+1}{\cos(\sin x)}$ is an even function. → f

4. If $\boxed{f(x+y) = f(x) \cdot f(y)}$ and $g(x) = \frac{f(x)}{1+(f(x))^2}$, then $g(x)$ is an even function. → T

   $f(x) = a^{kx}$

   $g(x) = \frac{a^{kx}}{1+a^{2kx}}$

5. Even functions are always many-one. → T

6. Odd functions are always one-one. → f

LAST CLASS REVISION

Answer the following Questions:

7. If $\boxed{f(x) = \frac{ax+b}{cx-a}}$, then $f(f(x)) =$ x , $f = f^{-1}$

8. If $\boxed{f(a-x) = f(a+x)}$, then graph is symmetrical about line $x = a$

9. If $f(a - x) = -f(a + x)$, then graph is symmetrical about point $(a, 0)$

10. If $\boxed{h(x) = f(x) - 1}$ and $\boxed{f(x) + f(1-x) = 2}$ $\forall x \in R$, then $\boxed{h(x)}$ is symmetrical about __________

point $\left(\frac{1}{2}, 0\right)$

(10)

$h(x) = f(x) - 1 \quad , \quad f(x) + f(1-x) = 2$

$x \to 1-x$

$h(1-x) = f(1-x) - 1$

$\underbrace{f(x) - 1} + \underbrace{f(1-x) - 1} = 0$

$h(x) + h(1-x) = 0$

$h(1-x) = \ominus h(x)$

$\boxed{x \to \frac{1}{2} + x}$

$h\left(1 - \frac{1}{2} - x\right) = -h\left(\frac{1}{2} + x\right)$

$h\left(\frac{1}{2} - x\right) = \ominus h\left(\frac{1}{2} + x\right)$

a a

Question Practice

Bumper Question

Check for following functions as Even/Odd/NONE.

(viii) $f(x) = \dfrac{x}{e^x - 1} + \dfrac{x}{2} + 1$ ⭐⭐⭐⭐⭐

Old IIT-JEE

$\boxed{x \to -x}$

$$f(-x) = \frac{-x}{\frac{1}{e^x} - 1} - \frac{x}{2} + 1$$

$$= \frac{x(1 - e^x - 1)}{1 - e^x} - \frac{x}{2} + 1$$

$$= x\left(\frac{1 - e^x}{1 - e^x} - \frac{1}{1 - e^x}\right) - \frac{x}{2} + 1$$

$$f(-x) = x + \frac{x}{e^x - 1} - \frac{x}{2} + 1$$

$$f(-x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1 = f(x).$$

Bumper Question

Check for following functions as **Even/Odd/NONE.**

(ix) odd. Function f satisfying : $\boxed{f(x+y)+f(x-y)=2f(x)\cdot f(y)}$ for all real values of x & y. Also f(0) = 0.

$f(-x)=f(x)$

$f(-x)=-f(x)$

$x=0$

$\boxed{f(y)+f(-y)=2\cancel{f(0)}\cdot f(y)}$

0

$\boxed{f(-y)=-f(y)}$

(x) $f(x) = x, \quad x \in Q$
$= -x, \quad x \in \overline{Q}$ odd.

DIBY-A

(xi) $f(x) = \dfrac{(3^x+1)^2}{3^x}$

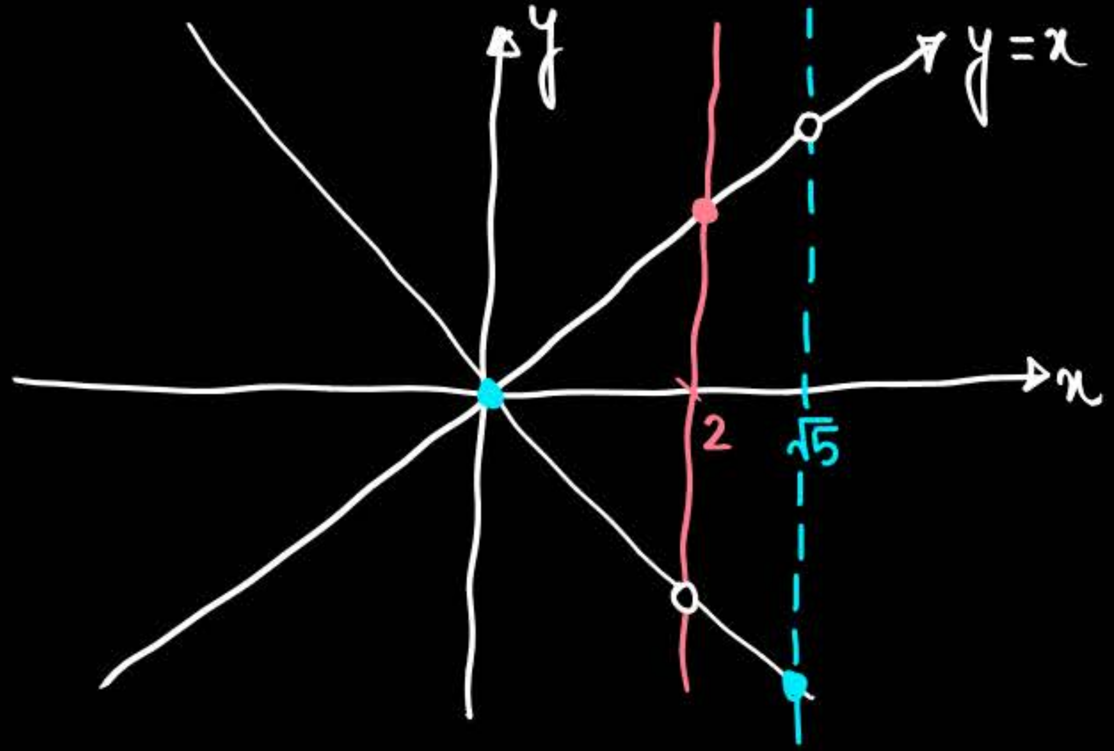
(xii) $f(x) = \ln\left(\dfrac{3-x}{3+x}\right)$

DIBY-B

(x) $\boxed{\begin{aligned} f(x) &= x \quad , x \in Q \\ &= -x \quad , x \in \overline{Q} \end{aligned}}$ "Tubelight"

Drichlet's func^n.

$$\left.\begin{array}{ll} f(2) = 2 & f(\sqrt{3}) = -\sqrt{3} \\ f(-2) = -2 & f(-\sqrt{3}) = \sqrt{3} \ldots\ldots \end{array}\right\} \text{odd.}$$



Question [JEE (Adv.)-2014]

$(\sec x - \tan x)(\sec x + \tan x) = 1$.

Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then

$x \to -x$

(A) $f(x)$ is an odd function

(B) $f(x)$ is a one-one function

(C) $f(x)$ is an onto function

(D) ~~$f(x)$ is an even function~~

$$f(-x) = \left(\log(\sec x - \tan x)\right)^3.$$

$$= \left(\log\left(\frac{1}{\sec x + \tan x}\right)\right)^3 = \left(-\log(\sec x + \tan x)\right)^3$$

$$= -f(x)$$

$$f(x) = \left(\log\left(\sec x + \tan x\right)\right)^3$$

*** $$\frac{dy}{dx} = 3\left(\log\left(\sec x + \tan x\right)\right)^2 \times \frac{1}{\cancel{(\sec x + \tan x)}} \times \overbrace{\cancel{(\sec x \tan x + \sec^2 x)}}^{\sec x}$$

$$\boxed{\frac{dy}{dx} = 3\left(\log\left(\sec x + \tan x\right)\right)^2 \sec x}$$

$, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

I & IV

$3 \to$ +ve, $\log(\sec x + \tan x) \to +$, $\sec x \to +$

$\frac{dy}{dx} \to +$ always $\Rightarrow f(x) \uparrow$

$\left(x \to \frac{\pi}{2}\right) \to f(x) = \left(\log\left(\sec x + \tan x\right)\right)^3 \to \infty$

$x \to 89.99999^\circ$

$\log \to \infty$, $\sec x \to +\infty$, $\tan x \to +\infty$

$$\# \left(x \to \frac{\pi}{2}\right) \longrightarrow f(x) = \Big(\log\big(\underbrace{\sec x}_{+\infty} + \underbrace{\tan x}_{+\infty}\big)\Big)^3 \longrightarrow \infty$$

$\hookrightarrow 89.99999^\circ$

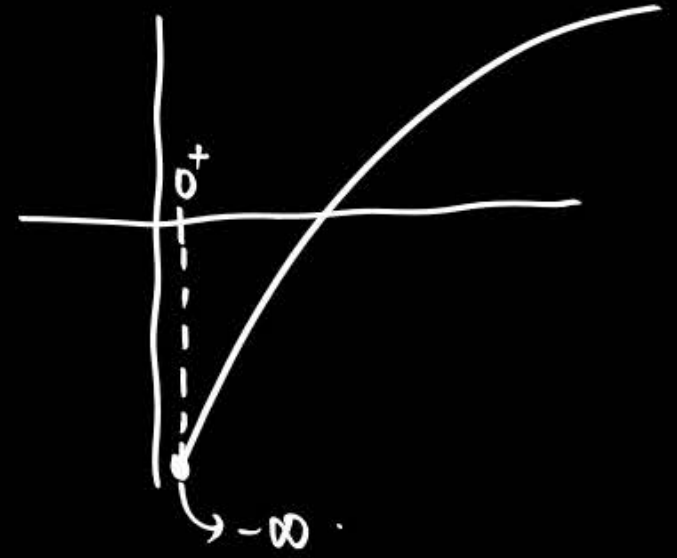
Range $\in \mathbb{R}$

$$\# \left(x \to -\frac{\pi}{2}\right) \longrightarrow f(x) = \Big(\log\big(\underbrace{\sec x}_{+\infty} + \underbrace{\tan x}_{-\infty}\big)\Big)^3 \longrightarrow -\infty$$

$\hookrightarrow -89.99999^\circ$

$$f(x) = \left(\log\left(\frac{1}{\underbrace{\sec x - \tan x}_{+\infty - (-\infty)}}\right)\right)^3 = \left(\log(0^+)\right)^3 = (-\infty)^3 = -\infty$$

$$\frac{1}{\infty} = 0^+$$



$\boxed{\log(\sec x + \tan x)}$ & $\boxed{\log\left(x + \sqrt{1+x^2}\right)}$

2 Bhai dono tabahi.

- odd
- 1-1.
- Onto

Question

Which of the following is not an odd function?

(A) odd. $g(x) - g(-x)$

(B) odd. $\left(g(x) - g(-x)\right)^3$

(C) $\log\left(\dfrac{x^4 + x^2 + 1}{x^2 + x + 1}\right)$

(D) odd. $xg(x) \cdot g(-x) + \tan(\sin x)$

$f(x) = g(x) - g(-x)$

$x \to -x$

$f(-x) = g(-x) - g(x)$

$= -(g(x) - g(-x))$

$= -f(x)$.

PROPERTIES OF EVEN-ODD FUNCTIONS

$f(x)$	$g(x)$	$f(x) + g(x)$	$f(x) - g(x)$	$f(x) \cdot g(x)$	$f(x)/g(x)$	$g(f(x))$	$f(g(x))$
Odd	**Odd**	odd	odd.	even	even	odd	odd.
Even	**Even**	even	even	even	even	even	even.
Odd	**Even**	NONE.	NONE.	odd.	odd.	even	even
Even	**Odd**	NONE.	NONE.	odd	odd.	even	even

Acidic Question

$$2\left[2+\frac{x}{\pi}\right]-3 = 2\left(2+\left[\frac{x}{\pi}\right]\right)-3 = 1+2\left[\frac{x}{\pi}\right]$$

Identify $f(x) = \dfrac{2x^5(\sin x + \tan x)}{2\left[2+\frac{x}{\pi}\right]-3}$, where [.] is G.I.F. as odd or even function.

$2\left[2+\frac{x}{\pi}\right]-3 \to 1+2\left[\frac{x}{\pi}\right]$

$x \to -x$

$$f(-x) = \frac{2(-x)^5\left(\sin(-x)+\tan(-x)\right)}{1+2\left[-\frac{x}{\pi}\right]} = \frac{2x^5(\sin x+\tan x)}{1+2\left[-\frac{x}{\pi}\right]}$$

$[x]+[-x] = 0, \ x \in I$

$= -1, \ x \notin I$

OR

$[-x] = -[x], \ x \in I$

$= -1-[x], \ x \notin I$

$\left[-\frac{x}{\pi}\right] = -\left[\frac{x}{\pi}\right], \ \frac{x}{\pi} = I \to x = I(\pi)$

$= -1-\left[\frac{x}{\pi}\right], \ \frac{x}{\pi} \neq I \to x \neq I(\pi)$

$x = I\pi$

$\# \ \boxed{f(-x) = 0}$

$x \neq I\pi$

$$f(-x) = \frac{2x^5(\sin x+\tan x)}{1+2\left(-1-\left[\frac{x}{\pi}\right]\right)} = \frac{2x^5(sx+tx)}{-\left(1+2\left[\frac{x}{\pi}\right]\right)} = -f(x)$$

$1+2\left(-1-\left[\frac{x}{\pi}\right]\right) \to 1-2-2\left[\frac{x}{\pi}\right] = -1-2\left[\frac{x}{\pi}\right]$

$f(-x) = \dfrac{2x^5(\sin x + \tan x)}{1 + 2\left[-\dfrac{x}{\pi}\right]}$

odd

$x = I\pi$

$x \neq I\pi$

$\boxed{f(-x) = 0}$

odd as well as even

$\boxed{f(-x) = -f(x)}$

odd.

Question

Prove that $f(x)=\frac{2(e^{x}-e^{-x})(\sin x+\tan x)}{2\left[\frac{x+2\pi}{\pi}\right]-3}$ is an $\boxed{\text{odd function}}$ where $[x]$ denotes the greatest integer function.

QIBY !!

Question

$f(x) = \underbrace{\cos x}_{\text{even}}$

If k is constant. Find the range of values of k if the function f, given by $f(x) = \left[\frac{x^2}{k}\right]\sin x + \cos x, -5 \leq x \leq 5$, is $\boxed{\text{even.}}$ (where [ ] denotes the greatest integer function)

$\left[\frac{x^2}{k}\right] \to 0$

$x \to (-x)$

$f(-x) = \boxed{-\left[\frac{x^2}{k}\right]}\sin x + \cos x$ $\to 0$

If $f(x)$ is even $\Rightarrow \left[\frac{x^2}{k}\right] = 0$, $x^2 \to [0, 25]$

$x \in [-5, 5]$

$x^2 \in [0, 25]$

$\boxed{K > 25}$

$0 \leq \frac{x^2}{K} < 1$

$K \in (25, \infty)$

$0 \leq x^2 < K$, $x^2 \to$ max.

$25 < K$

Question

odd

odd

$$f(x) = \ln\left(x^3 + \sqrt{x^6+1}\right) + \sin 5x + \left[\frac{x^2}{k}\right], \quad -3 < x < 3$$

"odd"

2 ×

1 ×

0

$x^2 \in [0, 9)$

$k \geq 9$

OR

$k \in [9, \infty)$

Question

Suppose that $f(x)$ is a function of the form $f(x) = \frac{ax^8+bx^6+cx^4+dx^2+15x+1}{x}$ $(x \neq 0)$. If $f(5) = 2$, then the value of $f(-5)$ is equal to

$f(-5) = 28$

$\overbrace{f(5)}^{2} + f(-5) - 30 = 0$

$\uparrow x = 5$

$f(x) + f(-x) - 30 = 0$

Odd funcⁿ

$\# \quad f(x) - 15 = \boxed{ax^7 + bx^5 + cx^3 + dx + \frac{1}{x}}$

$f(x) - 15 = g(x) \quad \text{——} \quad (1)$

$x \to -x$

$f(-x) - 15 = g(-x)$

$f(-x) - 15 = -g(x) \quad \text{——} \quad (2)$

EVEN-ODD EXTENSIONS

Let $f(x) = x + e^x + \sin x$ be defined on $[0, 2]$. Then find its even and odd extension.

$$f(x) = x + e^x + \sin x, \quad x \in [0, 2]$$

$f(-x)$

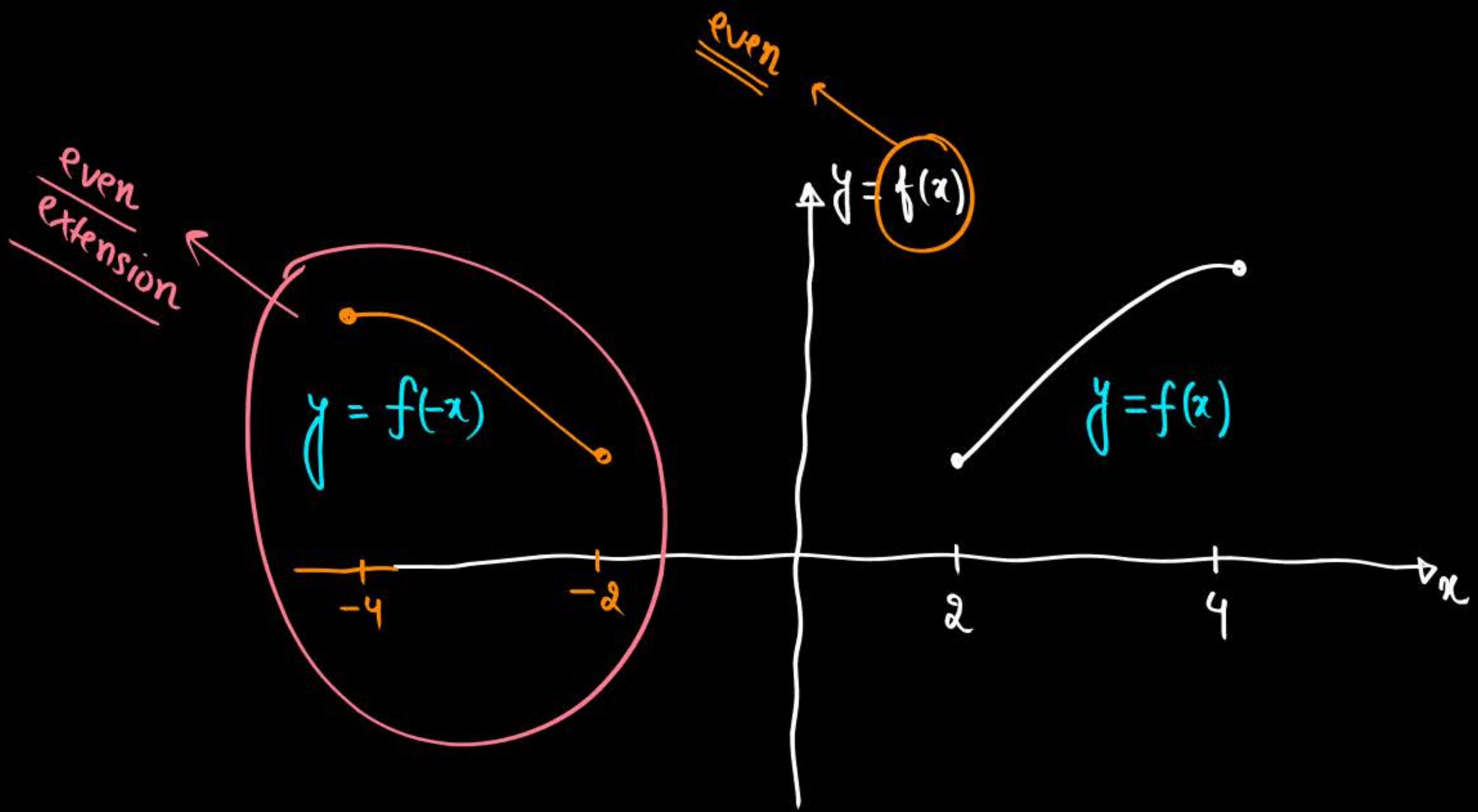
$$= \boxed{-x + e^{-x} - \sin x}, \quad x \in [-2, 0]$$

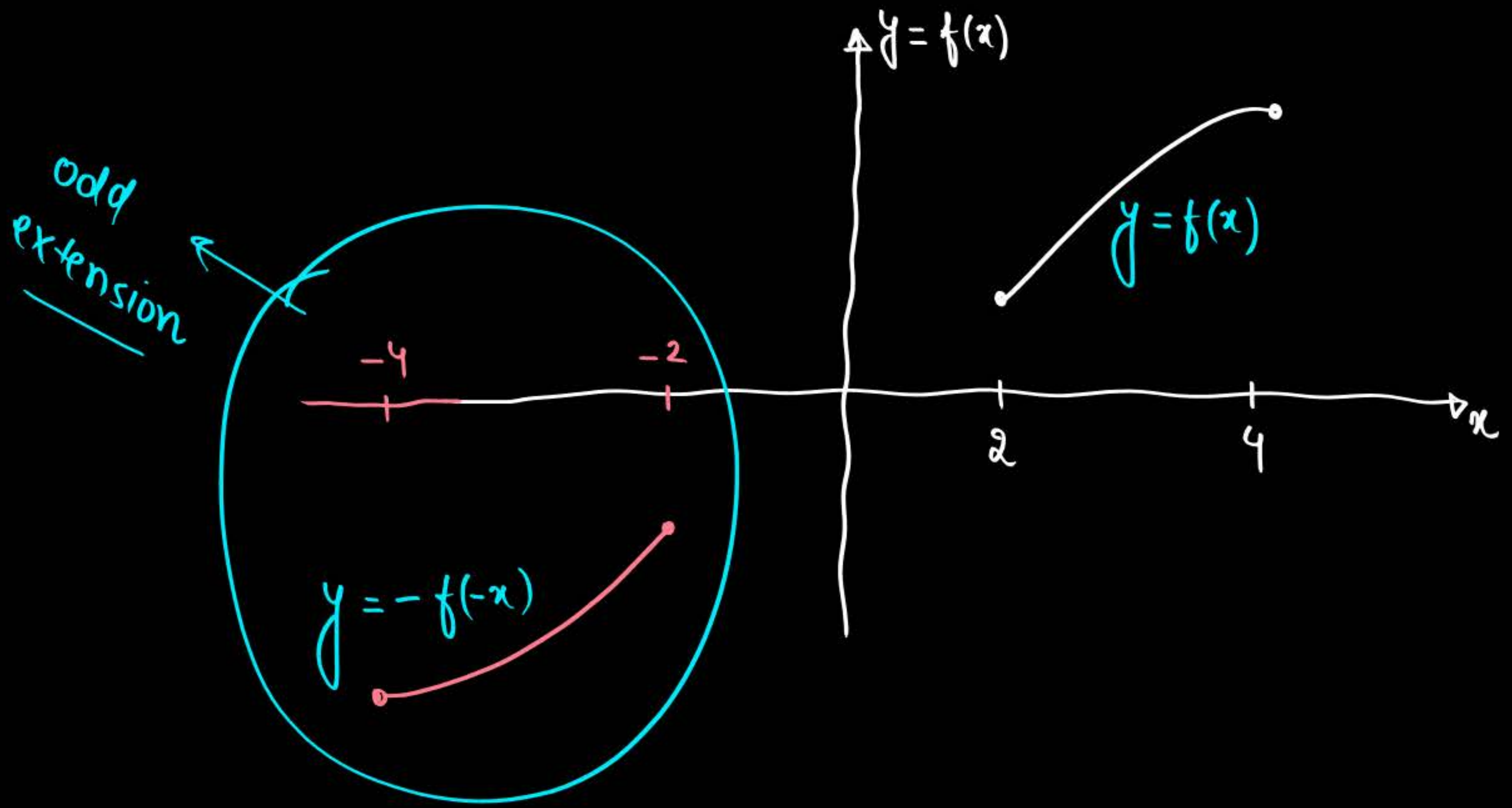
$$= \boxed{x - e^{-x} + \sin x}, \quad x \in [-2, 0]$$

$-f(-x)$

even

odd


even
even extension
y = f(x)
y = f(-x)
y = f(x)
-4
-2
2
4
x



odd
extension
$y=f(x)$
$y=f(x)$
-4
-2
2
4
x
$y=-f(-x)$

DO IT BY YOURSELF (DIBY)

DIBY-190

Let $f: \mathrm{R} - \{3\} \rightarrow \mathrm{R} - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$. Let $g: \mathrm{R} \rightarrow \mathrm{R}$ be given as $g(x) = 2x - 3$. Then the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to

[JEE Mains-2021]

(A) 7 (B) 2 (C) 5 (D) 3

[Ans. : (C)]

DIBY-191

For some $a, b, c \in N$, let $f(x) = ax - 3$ and $g(x) = x^b + c, x \in R$. If $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$, then $(fog)(ac) + (gof)(b)$ is equal to

[JEE Mains-2023]

[Ans. : 2039]

DIBY-192

Inverse of $f(x)=\begin{cases} x, & x<1 \\ x^2, & 1\le x\le 4 \\ 8\sqrt{x}, & x>4 \end{cases}$, is

[**Ans.** : $f^{-1}(x)=\begin{cases} x, & x<1 \\ \sqrt{x}, & 1\le x\le 16 \\ \frac{x^2}{64}, & x>16 \end{cases}$]

DIBY-193

Prove that $f(x)=e^x-e^{-x}$ is invertible and also find inverse of $f(x)$.

[**Ans.** :$f^{-1}(x)=\ln\left(\frac{x+\sqrt{x^2+4}}{2}\right)$]

DIBY-194

If $f(x)=\frac{x|x|}{1+x^2}$ then show that $f^{-1}(x)=\mathrm{sgn}(x)\cdot\sqrt{\frac{|x|}{1-|x|}}, -1<x<1$.

DIBY-195

If $f:[1,\infty)\to[2,\infty)$ is given by $x+\frac{1}{x}$, then $f^{-1}(x)$ equals [IIT-JEE 2001]

(A) $\frac{x+\sqrt{x^2-4}}{2}$ (B) $\frac{x}{1+x^2}$ (C) $\frac{x-\sqrt{x^2-4}}{2}$ (D) $1+\sqrt{x^2-4}$

[Ans. : (A)]

DIBY-196

Suppose $f(x)=(x+1)^2$ for $x\geq -1$. If $g(x)$ is the function whose graph is reflection of the graph of $f(x)$ with respect to the line $y=x$, then $g(x)$ equals [IIT-JEE 2002]

(A) $-\sqrt{x}-1, x\geq 0$ (B) $\frac{1}{(x+1)^2}, x>-1$ (C) $\sqrt{x+1}, x\geq -1$ (D) $\sqrt{x}-1, x\geq 0$

[Ans. : (D)]

DIBY-197

Let f be a one-one function with domain $\{x,y,z\}$ and range $\{1,2,3\}$. It is given that exactly one of the following statements is true and the remaining two are false $f(x)=1, f(y)\neq 1, f(z)\neq 2$ determine $f^{-1}(1)$. [IIT-JEE 1982]

[Ans. : y]

DIBY-198

Let $f(x)=\frac{ax}{x+1}, x\neq-1$. Then, for what value of a is $f(f(x))=x$? [IIT-JEE 2001]

(A) $-\sqrt{2}$ (B) $\sqrt{2}$ (C) 1 (D) -1

[Ans. : (D)]

DIBY-199

If $f(x)=\sin x+\cos x, g(x)=x^2-1$, then $g\{f(x)\}$ is invertible in the domain

[IIT-JEE 2004]

(A) $\left[0,\frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{4},\frac{\pi}{4}\right]$ (C) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (D) $[0,\pi]$

[Ans. : (B)]

DIBY-200

Let function $f: R\rightarrow R$ be defined by $f(x)=2x+\sin x$ for $x\in R$. Then, f is

[IIT-JEE 2002]

(A) one-to-one and onto
(B) one-to-one but not onto
(C) onto but not one-to-one
(D) neither one-to-one nor onto

[Ans. : (A)]

DIBY-201

There are exactly two distinct linear functions______and______ which map $\{-1, 1\}$ on to $\{0, 2\}$. [IIT-JEE 1989]

[Ans. : $f(x) = \pm x + 1$]

Homework

Re-attempt all good Questions of Class

DIBY

It is never about end result
It's always about Journey

THANK You

#futureIITians